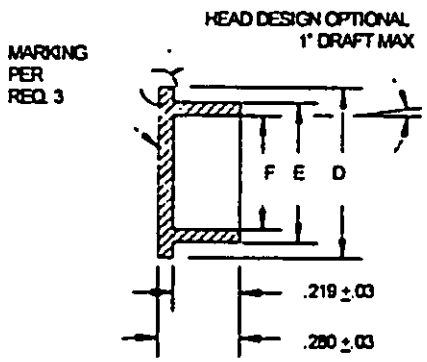
INCH-POUND
MIL-PRF-5501/32A
15 January 1997
SUPERSEDING
MIL-C-5501/32
29 July 1994

PERFORMANCE SPECIFICATION SHEET
CAP, "D" TYPE, ESD PROTECTIVE, ELECTRICAL CONNECTOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-C-5501 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.



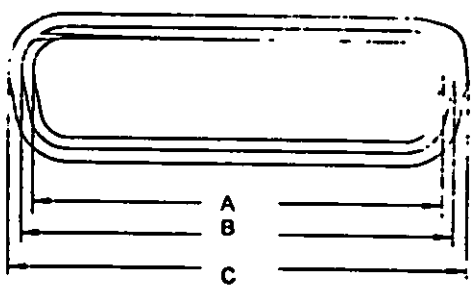


TABLE 1. "D" Type caps dimension.

DASH NO.	A ±.010	B ±.010	C ±.031	D ±.031	E ±.010	F ±.010
9S	.627	.694	.794	.462	.362	.295
9P	.708	.775	.849	.512	.434	.367
15S	.955	1.022	1.122	.462	.362	.295
15P	1.040	1.107	1.177	.512	.436	.369
25S	1.495	1.562	1.662	.462	.362	.295
25P	1.568	1.635	1.735	.530	.430	.363
37S	2.143	2.210	2.310	.462	.362	.295
37P	2.216	2.283	2.383	.530	.430	.363
50S	2.049	2.116	2.216	.574	.474	.407
50P	2.112	2.179	2.279	.636	.536	.469

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REQUIREMENTS:

1. MATERIAL - The cap shall be made of a permanent electrically conductive or static dissipative, non-sloughing material. Topical antistats shall not be used. The material shall have a volume resistance of less than or equal to 10^9 ohms measured as specified in note 2.
2. RESISTANCE TO SLOUGHING/CRAYONING - When rubbed briskly against a hard, flat laminated surface, the caps shall exhibit no tendency to slough or mark the surface in a manner similar to a crayon.
3. MARKING - Manufacturer name, EIA RS-471 electrostatic discharge sensitive symbol (hand in triangle) per MIL-STD-1686, and part number shall be marked on the outer face of the caps.
4. PART NUMBER - The part number consists of M5501/32A and a dash number taken from table 1 (example: M5501/32A - 15S).

NOTES:

1. Unless otherwise specified, dimensions are in inches.
2. Volume resistance shall be measured as follows:
    a. The test apparatus shall consist of an electrode, a calibrated ohmmeter and a non-oxidizing metal sheet. The electrode shall be constructed in accordance with ASTM F 150 with the following dimensional exceptions. The electrode and rubber insert diameter shall be 1.5 ±0.05 inches and the total weight shall be 2.0 lb ±0.1 oz. The ohmmeter shall be capable of supplying an open circuit test voltage of between 5 and 15 volts and shall be capable of measuring resistance of between 0 and 1,000,000,000 ohms. The non-oxidizing metal sheet shall be flat, free of any insulative coatings and at least 1/16 inch thick.
    b. All test samples shall be preconditioned at 10 ±2% RH and 70 ±2°F for 24 hours prior to testing in the same environment.
    c. Resistance measurements shall be made by laying the test cap on the metal sheet (open side up), filling the cap level with a conductive material such as foam or metal shot (#6 or finer), and placing the electrode on top. Ensure the electrode contacts both the filler material and the edge of the cap. Each measurement shall be made by connecting the leads of the ohmmeter to the top of the electrode and the metal plate and applying the test voltage. The resistance of the cap shall be as specified.
3. Sampling/Testing - One cap shall be selected at random per 1000 of each size purchased. When less than 1000 caps are purchased, one sample shall be selected at random for testing. For purchased caps having a dimension "A" of less than 1.5 inches, a cap from the same list having a dimension greater than 1.5 inches shall be used.
4. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.
5. Referenced Government (or non-Government) documents of the issue listed in that issue of the Department of Defense Index of Specification and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.
6. Electrically conductive or static dissipative material is defined as having a volume resistance of less than or equal to 10^9 ohms.

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AR
Navy - SH
Air Force - 11

Review activities:
Air Force - 70, 71, 80, 82, 84, 99
DLA - IS

Preparing activity:
Air Force - 11

(Project 5340-2347)

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